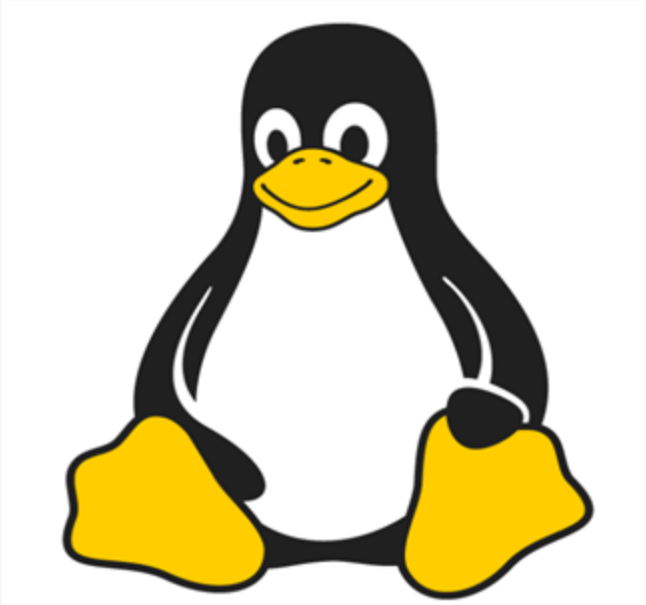
More Commands

LINUX OPERATING SYSTEM

- Directory structure
- Hard Link and Soft Link of file
- Metacharacters in Linux
- wc Command
- sort command
- head command
- tail command
- Redirections and pipes in Linux



Directory Structure in Linux ::

```
              /                              (root directory)
 --------------------------------
|       |        |          |
bin    sbin     dev     home            (every user has its own directory under home directory)
                                |
                  ----------------------------
                  |          |            |
                user1      user2        user3
```

. is a current dir (automatically created)

.. is a parent dir

~ :: Represents your home directory. (Directory in Linux means Folder in Windows sy.)

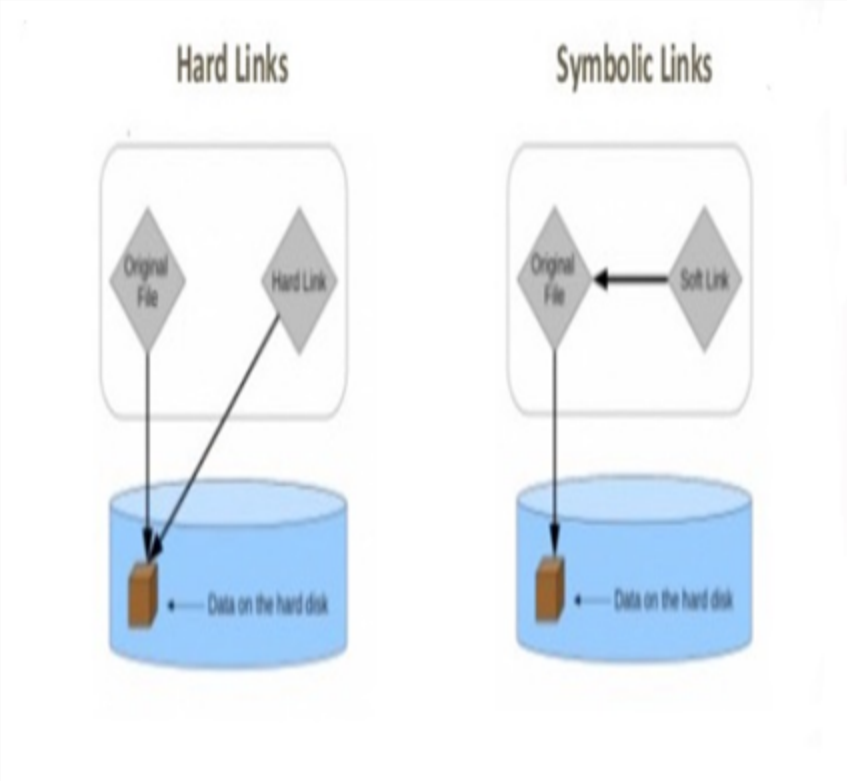
/ :: Represents root directory.

Hard Link and Soft Link of file

OBJECT™ TECHNOLOGIES

- **$ln f1 hdlnk**
  - This command creates a hardlink for file f1.
  - Here name of the hardlink is hdlnk.
  - This is a new file which physically exsists in the same directory .
  - Data in hdlnk will remain intact if f1 is deleted.
  - ls -l will display number of links for the file.

- **$ln -s f1 swlnk**
  - This command creates a softlink for file f1.
  - Name of the softlink is swlnk.
  - This is a new file which logically exsists in the same directory
  - If file f1 is deleted we will not get the data through this link.
  - ls -l will display the first bit as l for softlink.



Metacharacters in Linux

? Represents single character

* Represents set of characters

[ ] Represents set of characters
$ls [abc] will display all file names
starting with a or b or c.

[-] Represents rang of character set
$ls [a-d] will display all file names
starting from a or b or c or d.

Redirection and pipes

OBJECT™ TECHNOLOGIES

I/O Redirection ::

Linux has a special feature or special character which can redirect input or output whenever required.

Redirection here simply means diverting the output or input.

e.g $date > tempfile

Output of date is redirected to tempfile.

e.g $head < student.txt

File student.txt is given as a input to head command.

This is same as saying head studenr.txt

Overwrite

> standard output

< standard input

Append

>> standard output

<< standard input

Redirection and pipes

OBJECT™ TECHNOLOGIES

Piping ::

A pipe is a form of redirection (transfer of standard output to some other destination) that is used in Linux to send the output of one command/program/process to another command/program/process for further processing.

It is used to combine two or more commands, and in this, the output of one command acts as input to another command, and this command's output may act as input to the next command and so on.

It can also be visualized as a temporary connection between two or more commands/ programs/ processes.

```
command_1 | command_2 | command_3 | .... | command_N
```

e.g

```
$ ls  *.c  |  wc  -l
```

Here , output of first command is given as input to second command. Hence we get the count of .c files in current directory.

sort command

- $sort f1
  sorts the file.
- $sort -n f1
  sorting on numeric data
- $sort -k1 f1
  sort on 1st column
- $sort -k2 -n f1
  sort on second column which is numeric.
- $sort -k3 -nr f1
  sorting in reverse order.

nl command

- $nl filename
  allocates numbers to lines in file.

head command

- $head -n x filename
  displays first x number of lines from file.
- $head -n -x filename
  displays all lines from top excluding x lines from bottom
- $head -n +x filename
  same as head -n x filename

tail command

- $tail -n x filename
  displays all lines excluding last x lines.

- $tail -n -x filename
  same as above

- $tail -n +x filename
  displays all lines except (x-1) lines from the top.

find command

Find command is used to find some specific file in the given directory.

- $find dirname -options -command
  -name -print
  -type (f/d/l) -exec 'input'
  -size n,-n,+n

- $find . -type f -name "hello*"

- $find . -type f -empty

wc Command

- $wc filename

  without argument , command prints number of lines, words and bytes.

- $wc -l filename

  Prints the number of lines in a file.

- $wc -w filename

  Prints the number of words in a file.

- $wc -c filename

  Prints the count of bytes in a file.

Using pipes

- $ls | wc -l

- $find . -type d | wc -l

ps command

displays process status for any/all the commands.

- $ps –process status in current directory.
- $ps -e –all processess including system processess.
- $ps -e -o pid –e is all processess.
  –o is my option
- $ps -eo pid,ppid,%cpu,%mem –pid process id
  –ppid process parent id
  –%mem is memory utilization
  –%cpu is cpu utilization
- $ps -eo pid, %mem –sort==%mem –sorting the processess on memory utilization.
- $ps -eo pid, %mem –sort==-%mem –Sorting in reverse order.